Report about AdamW and Muon optimizers comparison

Kabanov D.S.

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Chapter 1

Task definition

Recently, the Muon optimizer, based on matrix orthogonalization, has demonstrated strong results in scaling the training process from small language models to larger models and achieving 2× computational efficiency compared to AdamW. The task is to perform a comparison of recent optimization strategies (AdamW, Muon) by **training time**, **maximum memory usage**, **final model quality**, and **training convergence**:

1. Download:
   - From Hugging Face: Qwen2.5-0.5B model, training dataset openwebtext-100k;
   - From GitHub: evaluation benchmarks (piqa, arc_easy, arc_challenge, winogrande, hellaswag) and MeZO optimizer if you want a challenge.
2. Perform **full fine-tuning** of the model with the following optimizers (using at least 1
   - AdamW;
   - Muon;
   - Improve Muon quality with a combination of Muon and AdamW. Model parameters are split into 2 groups. The first part of the model is trained with Muon, and the second part is trained with AdamW;
   - ***[Challenge]*** MeZO.
3. Compare **training speed**, **GPU memory usage**, **quality** of finetuned models and **training convergence**.
4. Present a report in LaTeX format.

Publish the solution as a public repository on GitHub. The report should be presented as a LaTeX document. Instructions should be presented in README.md.

Chapter 2

Algorithms description

2.1 AdamW

Adam optimizer (Adaptive Moment Estimation) — an algorithm for optimizing the weights of neural networks based on the backpropagation of error (gradient) and combining ideas from Nesterov Accelerated Gradient (***Momentum***) and RMSProp (***individual step for each parameter***). *The algorithm stores in memory two moments (matrices)* ***for each trainable parameter*** *of the model*:

1. The first moment (aka the moving average of the gradient, m_t) is an analogue of Momentum, which helps not to get stuck in local minima.

2. The second moment (aka the moving average of the square of the gradient, v_t) is an analogue of adaptive velocity, it helps to reduce the step where the gradients are very sharp and increase where they are flat.

Their combination allows the optimizer to take larger steps on flat terrain and smaller, careful steps on steep or "noisy" terrain.

Adam algorithm consists of the following steps (at each iteration t, the following calculations are performed for each parameter θ):

0) Initially, at $t = 0$, moments m_t (or m_0) and v_t (or v_0) for the weight matrices are zero.

1) The gradient g_t (the derivative of the loss function with respect to the current weight θ) is calculated using the formula:

$$g_t = \nabla_\theta L(\theta_t)$$

If L2-regularization is applied (λ is the regularization coefficient, weight decay), then the gradient of g_t is considered as:

$$g_t = \nabla_\theta L(\theta_t) + \lambda \cdot \theta_t$$

2) The moments are updated (using exponential attenuation $\sim$ so that the "old" gradients are forgotten and the "new" ones have more weight):

$$m_t = \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$$

$$v_t = \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2$$

β_1 (usually 0.9) determines how much we trust past experience. β_2 (usually 0.999) is responsible for the "memory" of gradient volatility.

3) Bias Correction. Due to the fact that at the very beginning of training m_t and v_t are equal to 0, they "accelerate" for a long time and tend to zero. To fix this, they are scaled (due to which, for small $t > 0$, their values are several times larger than they should be, but as t increases, β^t decreases, and therefore the scaling factor tends to 1):

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

4) After scaling, the model weights are updated (η — learning rate, ϵ — minimum addition so that there is no zero in the denominator):

$$\theta_{t+1} = \theta_t - \eta \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$$

---

input : γ (lr), β_1, β_2 (betas), θ_0 (params), $f(\theta)$ (objective)
λ (weight decay), *amsgrad*, *maximize*, ϵ (epsilon)
initialize : $m_0 \leftarrow 0$ (first moment), $v_0 \leftarrow 0$ (second moment), $v_0^{max} \leftarrow 0$

---

for $t = 1$ **to** $\ldots$ **do**
  if *maximize* :
    $g_t \leftarrow -\nabla_\theta f_t(\theta_{t-1})$
  else
    $g_t \leftarrow \nabla_\theta f_t(\theta_{t-1})$
  if $\lambda \neq 0$
    $\boxed{g_t \leftarrow g_t + \lambda \theta_{t-1}}$
  $m_t \leftarrow \beta_1 m_{t-1} + (1 - \beta_1) g_t$
  $v_t \leftarrow \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$
  $\widehat{m_t} \leftarrow m_t / (1 - \beta_1^t)$
  if *amsgrad*
    $v_t^{max} \leftarrow \max(v_{t-1}^{max}, v_t)$
    $\widehat{v_t} \leftarrow v_t^{max} / (1 - \beta_2^t)$
  else
    $\widehat{v_t} \leftarrow v_t / (1 - \beta_2^t)$
  $\theta_t \leftarrow \theta_{t-1} - \gamma \widehat{m_t} / (\sqrt{\widehat{v_t}} + \epsilon)$

---

return θ_t

The difference between AdamW and Adam lies in the place of application of L2 regularization (weight decay). When standard Adam applies regularization directly to gradients (see step 1 of the calculation, which can lead to undesirable effects), ***AdamW does this separately, at the parameter update level***, and not at the gradient level. In this way, AdamW implements the weight decay concept more accurately, which can lead to better learning outcomes. In other words, an additional step is added to AdamW (replacing the moment with L2-regularization).

The AdamW algorithm:

0) Initially, at $t = 0$, moments m_t (or m_0) and v_t (or v_0) for the weight matrices are zero.

1) The gradient g_t (the derivative of the loss function with respect to the current weight θ) is calculated using the formula:

$$g_t = \nabla_\theta L(\theta_t)$$

2) The moments are updated (using exponential attenuation $\sim$ so that the "old" gradients are forgotten and the "new" ones have more weight):

$$m_t = \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$$

$$v_t = \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2$$

β_1 (usually 0.9) determines how much we trust past experience. β_2 (usually 0.999) is responsible for the "memory" of gradient volatility.

3) Bias Correction. Due to the fact that at the very beginning of training m_t and v_t are equal to 0, they "accelerate" for a long time and tend to zero. To fix this, they are scaled (due to which, for small $t > 0$, their values are several times larger than they should be, but as t increases, β^t decreases, and therefore the scaling factor tends to 1):

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

4) **Regularization is applied to weights** θ (λ is the regularization coefficient, weight decay):

$$\theta_{t+1} = \theta_t - \eta \cdot \lambda \cdot \theta_t$$

5) The stage of updating the model weights (η — learning rate, ϵ — minimum addition so that there is no zero in the denominator):

$$\theta_{t+1} = \theta_{t+1} - \eta \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$$

Steps 4 and 5 can be rewritten as follows:

$$\theta_{t+1} = \theta_t - \eta \cdot \lambda \cdot \theta_t - \eta \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} = \theta_t - \eta \cdot \left(\frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} + \lambda \cdot \theta_t \right)$$

input : γ(lr), β_1, β_2(betas), θ_0(params), $f(\theta)$(objective), ϵ (epsilon)
λ(weight decay), *amsgrad*, *maximize*
initialize : $m_0 \leftarrow 0$ (first moment), $v_0 \leftarrow 0$ (second moment), $v_0^{max} \leftarrow 0$

for $t = 1$ **to** $\ldots$ **do**
  if *maximize* :
    $g_t \leftarrow -\nabla_\theta f_t(\theta_{t-1})$
  else
    $g_t \leftarrow \nabla_\theta f_t(\theta_{t-1})$
  $\boxed{\theta_t \leftarrow \theta_{t-1} - \gamma\lambda\theta_{t-1}}$
  $m_t \leftarrow \beta_1 m_{t-1} + (1-\beta_1) g_t$
  $v_t \leftarrow \beta_2 v_{t-1} + (1-\beta_2) g_t^2$
  $\widehat{m_t} \leftarrow m_t / (1-\beta_1^t)$
  if *amsgrad*
    $v_t^{max} \leftarrow \max(v_{t-1}^{max}, v_t)$
    $\widehat{v_t} \leftarrow v_t^{max} / (1-\beta_2^t)$
  else
    $\widehat{v_t} \leftarrow v_t / (1-\beta_2^t)$
  $\theta_t \leftarrow \theta_t - \gamma \widehat{m_t} / (\sqrt{\widehat{v_t}} + \epsilon)$

return θ_t

Advantages of the AdamW optimizer:

- Each parameter gets its own effective Learning Rate.
- Due to inertia (m_t), the algorithm wags less on noisy data.
- Adam usually converges faster than regular SGD and is less sensitive to the initial learning rate setting.
- Adam almost always works out of the box with default hyperparameters.

2.2 Muon

The **Muon** Optimizer (Momentum Updated Orthogonalized Optimizer) is an algorithm for optimizing the weights of neural networks based on orthogonalization of gradients (bringing the gradient matrix to an orthogonal form, where each column and row are orthogonal (perpendicular) to each other and have a length of 1). While standard optimizers like Adam adapt the step for each weight separately (using the moving average of the square of the gradient), Muon works with entire parameter matrices, trying to make them "as informative as possible", without duplicating information and preserving the spectral properties of the weight matrices.

Operating principle: Ordinary gradients often contain redundancy, due to which many directions in the weight space are updated almost identically. Muon applies an operation to the gradient matrix that makes it orthogonal (or approximately orthogonal). Muon, like Adam, also uses Momentum (m_t, the moving average of the gradient), which helps to avoid getting stuck in local minima. But, unlike Adam, Muon does not store the moving average of the squares of the gradient to adapt the step, which reduces its memory requirements.

The Muon algorithm:

0) Initially, at $t = 0$, the Momentum m_t (or m_0) of the weight matrices is zero.

1) The gradient g_t (the derivative of the loss function with respect to the current weight θ) is calculated using the formula:

$$g_t = \nabla_\theta L(\theta_t)$$

2) Consideration of Momentum, which makes it possible to smooth out noise and keep the inertia of movement (in a different way than in Adam) to a minimum of the loss function (coefficient μ, in Adam it is $\beta_1 \sim$ how much do we trust past experience):

$$m_t = \mu \cdot m_{t-1} + g_t$$

If we use Nesterov's Momentum, then:

$$m_t = \mu^2 \cdot m_{t-1} + (1 + \mu) \cdot g_t$$

3) Calculation of orthogonalization of the gradient matrix O_t, for which Muon takes the matrix of accumulated gradients and applies the Newton-Schultz iterative procedure to it to obtain an approximation (instead of the classical SVD decomposition, which is computationally expensive):

$$O_t = NS_k^{(a,b,c)}(m_t; \epsilon)$$

Here (a, b, c) are the coefficients of the Newton-Schultz orthogonalization polynomial, ϵ is the minimum addition so that there is no zero in the denominator.

Due to this, the product of the matrix itself is $OO^T \approx 1$, all singular values of the matrix become equal to 1, which removes the problem of different gradient scales for different layers and parameters.

4) **Appling L2 regularization to weights** (λ is the regularization coefficient or weight decay, η — initial learning rate):

$$\theta_{t+1} = \theta_t - \eta \cdot \lambda \cdot \theta_t$$

5) Recalculation the update step η (learning rate) depending on the dimension of the weight matrix (so that the orthogonalized update has a consistent RMS value for rectangular matrices) and taking into account the orthogonal gradient matrix for updating the weights:

$$\eta = AdjustLR(\eta; shape(\theta_t))$$

$$\theta_{t+1} = \theta_{t+1} - \eta \cdot O_t$$

Steps 4 and 5 can be rewritten as follows:

$$\theta_{t+1} = \theta_t - \eta \cdot \lambda \cdot \theta_t - AdjustLR(\eta; shape(\theta_t)) \cdot O_t$$

---

input : γ (lr), λ (weight decay), μ (momentum), $nesterov \in \{True, False\}$,
(a, b, c) (NS coefficients), ε (epsilon), k (NS steps), θ_0 (params), $f(\theta)$ (objective)
initialize : $B_0 \leftarrow 0$ (momentum buffer)

---

for $t = 1$ **to** $\ldots$ **do**
  $g_t \leftarrow \nabla_\theta f_t(\theta_{t-1})$
  $B_t \leftarrow \mu B_{t-1} + g_t$
  $\widetilde{B}_t \leftarrow \begin{cases} g_t + \mu B_t, & \text{if nesterov} = True \\ B_t, & \text{if nesterov} = False \end{cases}$
  $O_t \leftarrow \mathrm{NS}_k^{(a,b,c)}(\widetilde{B}_t;\ \varepsilon)$
  $\theta_t \leftarrow \theta_{t-1} - \gamma\, \lambda\, \theta_{t-1}$ (decoupled weight decay)
  $\gamma \leftarrow \mathrm{AdjustLR}(\gamma;\ \mathrm{shape}(\theta_t))$
  $\theta_t \leftarrow \theta_t - \gamma\, O_t$

---

return θ_t

Advantages of the Muon optimizer:

- Aligns the update scale in all singular directions (Muon forces all directions in the scale space to be trained at the same speed, whereas in Adam some directions can "fade out" faster than others).
- Accelerates convergence, as the optimizer does not waste time "rocking" in the narrow canyons of the loss function.
- Allows you to use a higher learning rate without losing stability.
- Requires less memory, as it stores only one moving average (for Momentum) instead of two.

Muon is designed specifically for **two-dimensional weight matrices** (linear layers), while AdamW is usually used for the rest of the parameters (embedding, normalization, bias).

Chapter 3

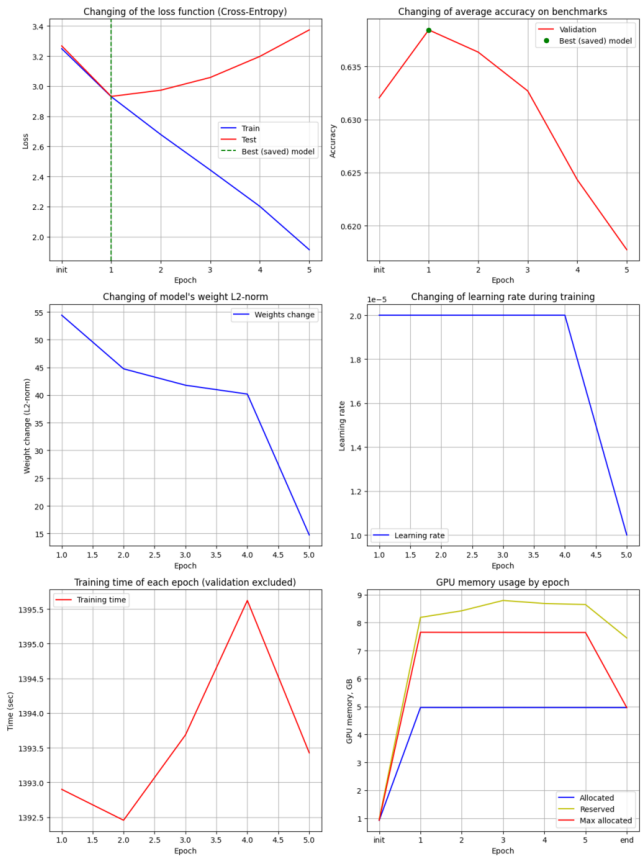
Fine-tuning results

For the correctness of the comparison, the following hyperparameters were set for all fine-tuning experiments:

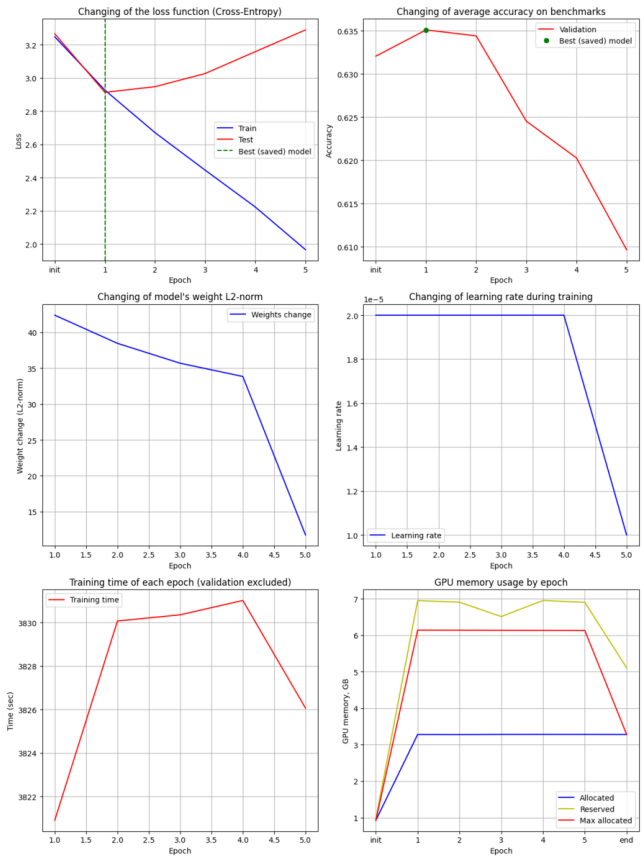
- Maximum number of samples: 4000
- Test proportion: 10%
- Additional validation during training on benchmarks: "piqa", "winogrande"
- Limit on the number of tokens generated by the model: 256
- Maximum number of training epochs: 20
- Epochs before early stopping of training (without improving the tracked metric and/or convergence of weights): 3
- Tracked metric for training early stopping: AVG Accuracy on benchmarks
- Initial learning rate: $2 \cdot 10^{-5}$
- Learning scheduler that changes the learning rate 0.5 times every 4 epochs
- L2 regularization coefficient to prevent overfitting: 0.1
- Momentum coefficient: 0.9
- Batch size: 4
- Random seed: 42

Changing some of hyperparameters can affect the training and convergence process both for the better (increasing the batch size for Muon acceleration, since it consumes less memory) and for the worse (reducing the learning rate, to which Muon is highly sensitive, will lead to a significant decrease in the steps of changing weights and, consequently, slowing down the updating of weights).

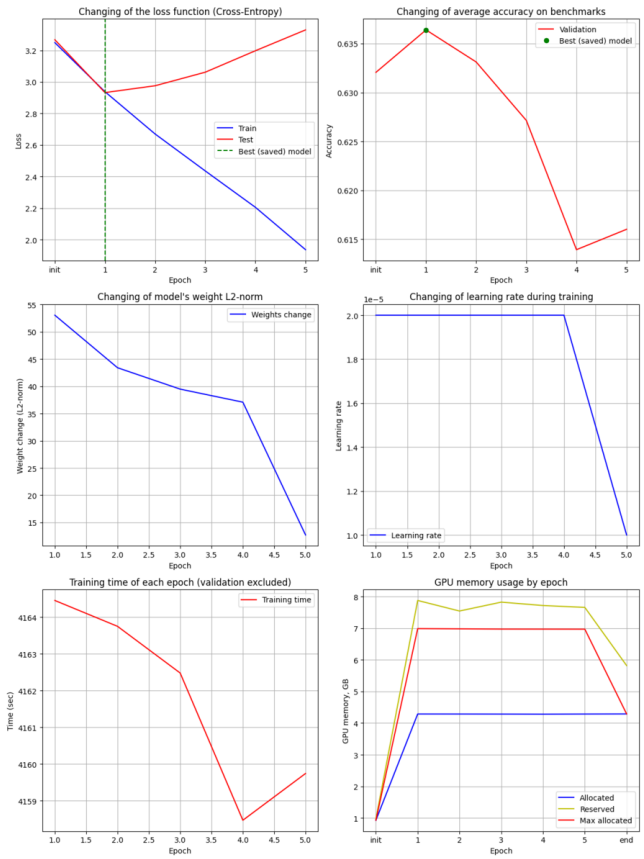
3.1 AdamW results



3.2 Muon results



3.3 Muon with AdamW results

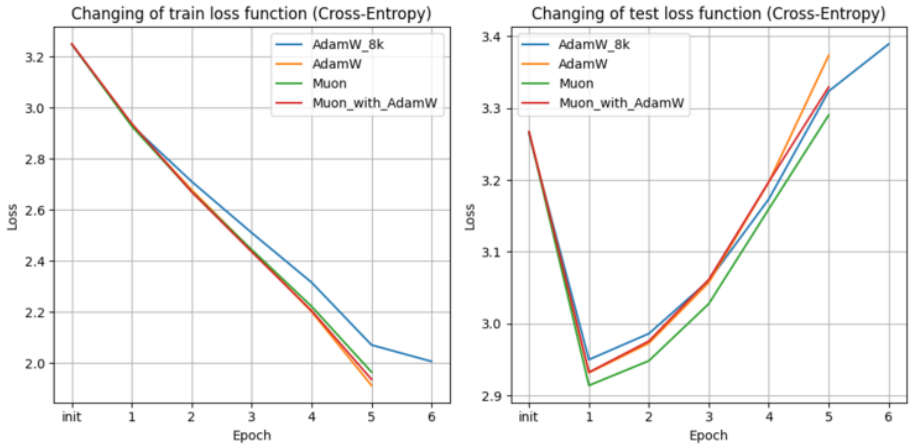


Chapter 4

Results comparison

For a more comprehensive analysis, another experiment was conducted, which included 8000 samples instead of 4000. The training was only with AdamW optimizer (*AdamW_8k*), since other optimizers simply did not fit into the Kaggle quota.

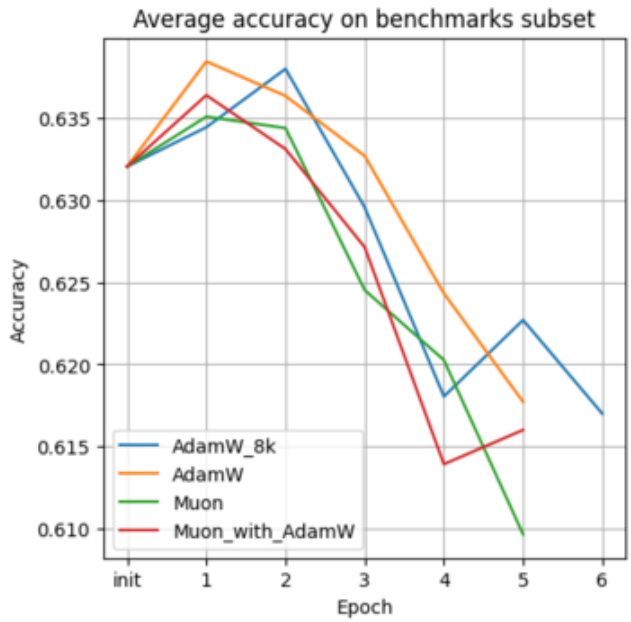
4.1 Cross-Entropy Loss



When fine-tuning with any optimizer (*including AdamW_8k experiment*), the **change in Cross-Entropy loss**, both on training data and on test data, has approximately the ***same trend. Fast learning with a strong loss decrease on train data, but at the same time overfitting after the first epoch*** (loss on test data has a V-shaped trend). This observation can be explained by

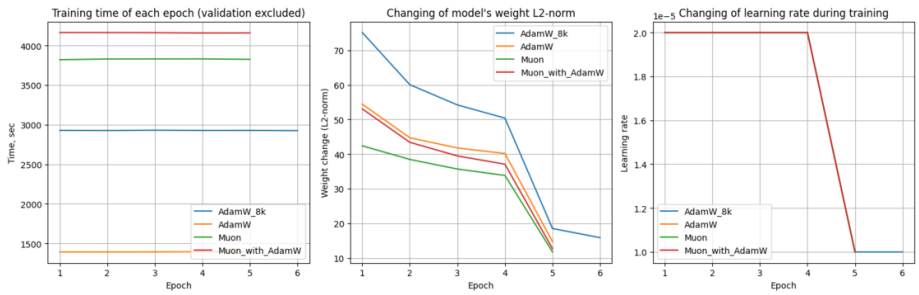
the fact that the model is too large for such a small amount of disparate data (even increasing to 8000 samples did not solve the problem).

4.2 Average accuracy on benchmarks subset



The **change in tracked metric** (*average accuracy on subset of benchmarks*) also shows a similar ***trend among all experiments, where the metric improves at 1-2 epochs, after which there is a sharp decline due to overfitting***. At the same time, training with AdamW (*AdamW, AdamW_8k, Muon_with_AdamW experiments*) shows slightly higher accuracy than when training only with the Muon optimizer (*it should be clarified that only 72% of the model weights are trained in Muon, whereas in other experiments all 100%*).

4.3 Training time and convergence



Training time varies greatly between optimizers, where:

1. ***AdamW optimizer are the fastest*** (even with twice the amount of input data), ∼1400 seconds per 4000 samples (23 minutes per epoch) and ∼3000 seconds (50 minutes per epoch with 8000 samples). The training time depends almost linearly on the amount of data.

2. ***Muon*** optimizer, even with only 4000 samples, spent ∼3,800 seconds per epoch, which is just over an hour and which is almost ***3 times worse than AdamW*** on the same data.

3. ***Combination of Muon with AdamW showed the longest training time*** (∼4100 seconds), since optimization of other weight matrices using AdamW was added on top to the weight optimization with Muon.

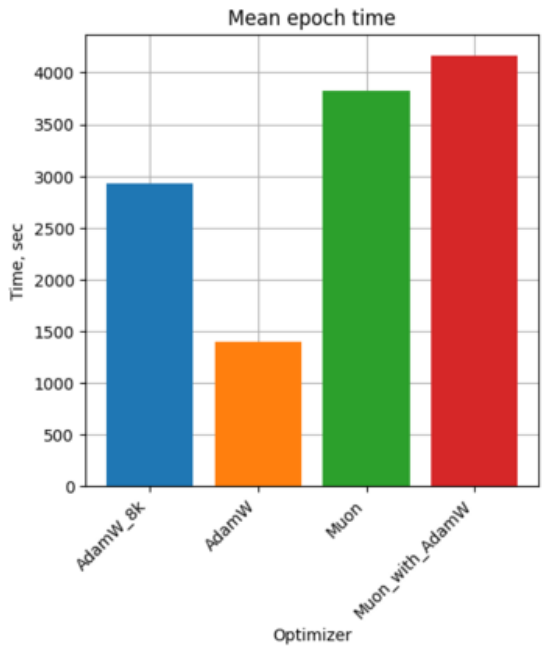
L2 norm of change in weights during training also varies.

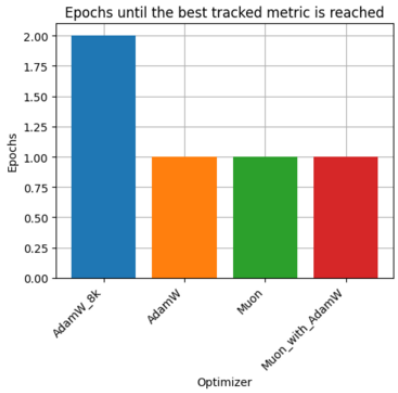
1. ***Weights changed the least when optimized by the Muon algorithm***, while the value of loss function was comparable to other optimizers. That is, to achieve similar "progress", it is enough for Muon optimizer to change a smaller number of weights (only 72% are trained, not 100%) and to a lower value (*while the loss on the test data is even better than that of other optimizers, which cannot be said about the accuracy on benchmarks*).

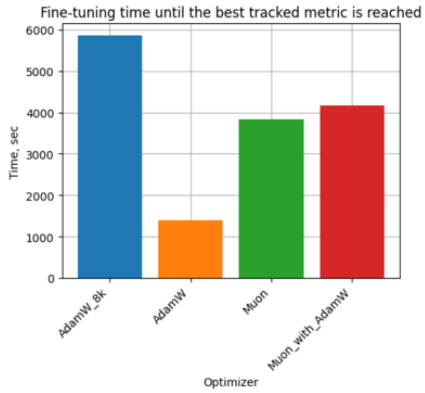
2. ***The weights changed the most when training with AdamW***, with a 25% more pronounced shift in relation to Muon.

3. Changing the weights with a ***combination of optimizers turned out to be something in between Muon and AdamW***.

It is also worth to note that with more data, the more the weights of the model changed (according to experiments with 4000 and 8000 samples).

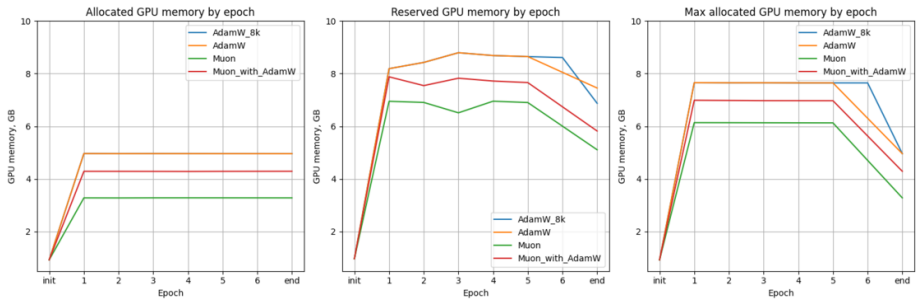




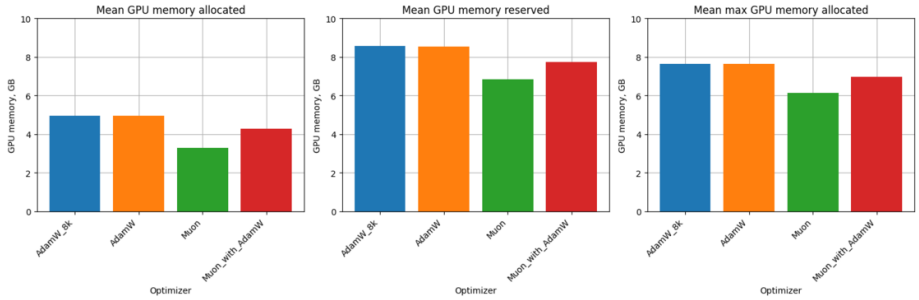


It took only one epoch for the models to be fine-tuned to the best accuracy on the benchmarks at 4000 samples. *If training on more data, then along with the increase of computing time, the accuracy will also be higher.*

4.4 GPU memory usage



Between epochs, GPU memory usage remains almost unchanged (for each optimizer), so proceed to consider the average value of GPU usage.



Memory usage depends only on the type of optimizer and does not depend on the dataset size (since only a fixed-size batch is loaded at one time).

GPU allocated memory after an epoch:

- ***The highest for the AdamW optimizer***, since it stores two moments at once for each trainable weights (the moving average of the gradients and the moving average of the squares of the gradients). It takes up 4.96 GB with the model.
- After one fine-tuning epoch, ***Muon frees up more memory***, by about 33%, compared to AdamW. Occupying only 3.28 GB with the model.
- ***Combination of Muon and AdamW has average consumption*** (4.29 GB), after training they take up 15% less memory compared to AdamW, and 30% more when compared with only Muon.

Reserved (cached) GPU memory:

- Similarly, AdamW optimizer reserves the most memory, 8.55 GB.
- One Muon reserves 20% less memory (6.84 GB).

- The combination of optimizers again has an intermediate value (7.73 GB), 10% less than AdamW, but 13% more than Muon.

Maximum GPU allocated memory during training epoch:

- ***AdamW has the highest***, 7.65 GB.
- ***Muon consumes 20% less GPU memory*** (6.13 GB) than AdamW during training.
- The optimizer combination consumes 6.98 GB, which is 9% less and 14% more than AdamW and Muon, respectively.

4.5 Benchmarks

Validation data is the following datasets that evaluate the understanding of the model of physical properties and logical relationships between objects, as well as the possibilities of commonsense reasoning:

1. PIQA (Physical Interaction: Question Answering) — benchmark for determining how well a language model understands the physical properties of objects (*fragility, hardness*) and everyday actions (*how best to wash dishes or place an object*).
2. ARC easy (AI2 Reasoning Easy) — benchmark created to evaluate the ability of artificial intelligence systems to answer questions that require basic scientific knowledge and logic. The set consists of multiple-answer questions taken from real school science exams (grades 3-9). The "Easy" category includes questions that modern (at the time of creation) models could answer using simple statistical methods or keyword search.
3. ARC challenge (AI2 Reasoning Challenge) — structurally similar to "ARC easy", but contains more difficult questions that require "multi-layered" reasoning to answer.
4. WinoGrande — benchmark designed to evaluate the ability of artificial intelligence models to reason based on common sense (commonsense reasoning). The benchmark is based on tasks like the Winograd Schema Challenge (WSC), where the model needs to correctly correlate a pronoun with one of two objects mentioned in the sentence. For example, there is an expression "The trophy did not fit in a brown suitcase because *** was too big." The model needs to choose correct answer from a suggested ones. Options: A) Trophy; B) Suitcase. The correct answer is: A (Trophy).
5. HellaSwag (Harder Entities, Longer Contexts, and Better Adversarial Filtering) — benchmark created to evaluate the commonsense reasoning abilities of large language models (LLM). The benchmark checks how well the model can predict the most logical outcome of an everyday scenario. Unlike simple tests, HellaSwag uses the Adversarial Filtering method: the

answer options are generated so that they look plausible to algorithms, but are obviously incorrect to humans. The model gets a context (for example, a description of the beginning of a video clip) and four continuation options. The task is to choose the right one. Tasks are chosen so that they are easy for humans (accuracy $\sim 95\%$), but for a long time remained extremely difficult for AI.

The main metric of all benchmarks is **Accuracy**, that is, the indexes of answers predicted by a model (0, 1, ...) are compared with true label. The higher the Accuracy, the better the model understands essence of the sentence. To do this, CausalLM model is used to estimate logarithmic probability (log-likelihood) of each of the response options. The model "chooses" option that is more likely to be generated with a given context (input).

	Baseline	AdamW_8k	AdamW	Muon	Muon_with_AdamW
arc_challenge	0.292662	0.302048	0.298635	0.289249	0.296075
arc_easy	0.646465	0.658670	0.659512	0.654461	0.661616
hellaswag	0.406094	0.413264	0.401514	0.400119	0.400020
piqa	0.702938	0.704570	0.707835	0.703482	0.702938
winogrande	0.561168	0.571429	0.569061	0.566693	0.569850

Comparing **Accuracy on benchmarks**, it can be seen that:

- ***The best metrics, as expected, were shown by experiment with the largest number of samples*** (8000 versus 4000 for all other optimizers). It surpassed the "baseline" (model without fine-tuning) in all benchmarks by 0.2 — 1.2% Accuracy.
- ***AdamW***, trained on 4,000 samples, ***ranks second*** in terms of Accuracy. He even managed to slightly (by $\sim$0.001 — 0.003, which is less than half a percent) overtake AdamW_8k on "arc_easy" and "piqa" benchmarks.
- ***The combination of optimizers showed the most unstable metrics***, for benchmarks "arc_easy" and "winogrande" there is a relatively good increase in Accuracy, while for "hellaswag" and "piqa" model performed worse than its untrained version.
- ***One Muon showed the worst results***, in two benchmarks it was worse than untrained model ("arc_challenge" and "hellaswag"), and in the other three it was only slightly better ("arc_easy", "piqa", "winogrande").

Chapter 5

Conclusion

Key Findings:

- The ***final quality of a model is more influenced by amount of data than a type of optimizer*** (*although fune-tuning with Muon showed almost no improvement in Accuracy*).
- The ***convergence of models with respect to the loss function is almost the same for optimizers***, but not according to L2 norm of updated model's weights, they change the most with AdamW optimizer, the weakest with Muon since not all parameters are trained (*while their loss is almost the same*).
- ***Muon is running much slower*** (*3 times*) than AdamW (*although this may be due to the architecture of a model*).
- ***Muon requires less memory*** (*by ∼20% compared to AdamW*), as it stores only one additional matrix for weights (Momentum) and it doesn't fine-tune all the parameters.
- There are perspectives for Muon related to increasing the batch size and learning rate.
- ***The combination of optimizers shows intermediate results*** in all analysis, except for training time, where fine-tuning took the longest.

Opportunities for further research:

- Using the MeZO optimizer;
- Training on full dataset;
- Using a model for which orthogonalization of matrices would be more profitable and faster than calculating gradients;
- Increase the learning rate and batch size.